

GeoSentinel AI — Analysis Report

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Analysis Period

Field	Value
Date 1 (T1)	2022-11-15
Date 2 (T2)	2024-01-15
T1 Scene ID	—
T2 Scene ID	—
T1 Acquisition	—
T2 Acquisition	—
T1 Cloud Cover	—
T2 Cloud Cover	—
Satellite	Sentinel-2 L2A
Resolution	10 metres
Processing Time	— s

Land Cover Area Change

Class	T1 Area (km²)	T2 Area (km²)	Change (km²)	Change (%)
Background	8.372	8.370	-0.002	-0.0%
Urban	0.000	0.000	+0.000	+0.0%
Vegetation	0.150	0.136	-0.013	-8.9%
Water	0.000	0.000	+0.000	+0.0%
Barren	0.225	0.241	+0.015	+6.8%
Agriculture	0.000	0.000	+0.000	+0.0%

Recommendations

[CRITICAL] Critical Vegetation Loss Detected

Why: Vegetation cover decreased by 44.3% (3.64 km²) between 2022-11-15 and 2024-01-15, which exceeds the critical threshold of 25%.

Action: A critical level of vegetation cover has been lost in the selected area. Immediate field investigation is recommended. Initiate emergency afforestation programs and restrict any further land clearing activities. Notify the Telangana Forest Department for intervention.

[HIGH] Significant Vegetation Loss Detected

Why: Vegetation cover decreased by 44.3% (3.64 km²) between 2022-11-15 and 2024-01-15, exceeding the high-alert threshold of 15%.

Action: Significant vegetation loss has been detected. Plan and execute targeted afforestation programs in affected zones. Identify drivers (construction, agriculture conversion, fire) through ground truth

verification. Monitor monthly for further changes.

[HIGH] Rapid Urban Expansion Detected

Why: Urban cover increased by 44.3% (3.64 km²) between 2022-11-15 and 2024-01-15, exceeding the rapid-expansion threshold of 10%.

Action: Rapid urbanization is occurring in the study area. Engage HMDA (Hyderabad Metropolitan Development Authority) to review Master Plan compliance. Prioritize green buffer zones and ensure infrastructure (roads, water, drainage) is upgraded proportionally.

[HIGH] Land Cover Change Hotspots Identified

Why: 3 change hotspot(s) were detected, each exceeding 0.5 km². The largest hotspot covers 0.01 km² centered at 17.3850°N, 78.4867°E.

Action: Concentrated change hotspots have been identified. These may indicate targeted development activity or environmental disturbance. Dispatch field teams to highest-priority locations for immediate ground verification and documentation.